Group Assignment

Real World Application of Goal Programming

MSF1222-Operational Research II

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Application of Goal Programming Model for Profit Maximization of Real World Problem

CASE STUDY OF A BAKERY PRODUCTS COMPANY

FOOD ART (Pvt) Ltd

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Introduction

Goal programming is a mathematical optimization technique that allows for multiple, often conflicting objectives to be considered simultaneously. Unlike traditional linear programming, which focuses on maximizing or minimizing a single objective function, goal programming Aims to achieve a set of goals as closely as possible.

Goal programming is mainly use for solving real world problem such as Production planning, waste management, urban planning etc.

Here, we have choose a company and create a model to maximizing their profit. We have visit the company and collected the following data.

Company name: Food Art (pvt) Ltd

Food Art (pvt) Ltd is a company that make bakery items. According to their supplied data they are mainly supplied bread and buns for the local market. They want to maximize their profit by maximizing their production according to some priorities.

In this study, our aim is to find the best way to get the maximum weekly profit on the owner's hand. Therefore we have used Goal programming method to find the maximum number of breads and buns with their requirements and conditions.

Methodology

We have collected the following data by visiting the company.

Food Art (pvt) Ltd is mainly produce breads and buns. We collected data for daily and monthly basis and convert them to the weekly basis for easiness of solving.

They are normally produce 1500 breads and 9000 buns daily. The manager wants to maximize the production within some boundaries.

Working days per week - 7

Cost for the ingredients

Ingredients	Quantity(daily)	Unit price(1 kg)
Flour	550 kg	Rs.187
Margarine	35 kg	Rs.900
Sugar	120 kg	Rs.260
egg	300	Rs.37.50(1 egg)
salt	7.5 kg	Rs.100
Total daily cost	Rs.177 550	
Total weekly cost	**Rs.1 242 850**	

Other cost

Reason	Cost (Monthly)
Electricity bill	Rs.500 000
Water bill	Rs.100 000
Transportation	Rs.600 000
Salary	Rs.1 225 000
Total monthly cost	Rs.2 425 000
Total weekly cost	**Rs.606 250**

Total weekly cost=Rs.1 849 100

Working Hours

Employees are working two shifts for a day.

- Day shift- 6 a.m.-5 p.m.
- Night shift- 5 p.m-4 a.m.
- Overtime for a day- 2 hours

Item	Packing hours(daily)	Baking Hours(daily)
Bread	3 h	4 h
Bun	7 h	8 h
Total daily hours	10h	12 h
Total weekly hours	**70 h**	**84 h**

Production

Item	Packing(units per hour)	Baking (units per hour)
Bread	650	500
Bun	2000	1800

Problem

Food Art (pvt) Ltd is a company run on bakery products. The company manager wants to maximize the profit under some conditions. Below are the conditions according to the order of the priorities.

Goal 1 - profit should be at least Rs.1 400 000

Goal 2 – Total cost should be less than Rs.1 900 000

Goal 3 – Avoid the use of over time for packing

Goal 4 – Avoid the use of over time for baking

Find the amount of breads and buns in order to achieve the goals.

Formulation of the model

Goal Programming model

Objective: Maximize the total weekly profit

Decision Variables:

Let,

X_1 =Number of breads manufactured during a week

X_2 =Number of buns manufactured during a week

Goals:

Goal 1 - profit should be at least Rs.1 400 000 (P1)

Goal 2 – Total cost should be less than Rs.1 900 000 (P2)

Goal 3 – Avoid the use of over time for packing (P3)

Goal 4 – Avoid the use of over time for baking (P4)

- P denotes the priority levels.

Deviations:

d_1^+ - over achievement of goal 1

d_1^- - under achievement of goal 1

d_2^+ - over achievement of goal 2

d_2^- - under achievement of goal 2

d_3^+ - over achievement of goal 3

d_3^- - under achievement of goal 3

d_4^+ - over achievement of goal 4

d_4^- - under achievement of goal 4

Non negativity constraints:

$x_i , d_j^+ , d_j^- >= 0$

$i=1,2$ & $j=1,2,3,4$

- The goal programming model can be formulated as follows.

Objective function

Minimize $\{P_1d_1^- , P_2d_2^+, P_3d_3^+ , P_4d_4^+\}$

Subject to,

Goal 1 : $30x_1+10x_2+d_1^- -d_1^+=1\ 400\ 000$

Goal 2 : $90x_1+17x_2+d_2^- -d_2^+=1\ 900\ 000$

Goal 3 : $1/650x_1+1/2000x_2+ d_3^- -d_3^+=70$

Goal 4 : $1/500x_1+1/1800x_2+ d_4^- -d_4^+=84$

$x_i , d_j^+ , d_j^- >= 0$

$i=1,2$ & $j=1,2,3,4$

Solution

Here we have solve the goal programming model using matlab.

By using matlab

Matlab code

```
Aeq = [
    30 10 1 -1 0 0 0 0 0 0;
    90 17 0 0 1 -1 0 0 0 0;
    1/650 1/2000 0 0 0 0 1 -1 0 0;
    1/500 1/1800 0 0 0 0 0 0 1 -1;
    ];

% Right-hand side of the equality constraints
beq = [1400000; 1925000; 70; 84];

% Objective function coefficients (column vector)
f = [0; 0; 1; 0; 0; 1; 0; 1; 0; 1]; % Coefficients in objective
function

% Lower bounds for the variables (all must be non-negative)
lb = zeros(10,1);

% Upper bounds (optional, set to inf)
ub = [];

% Solve the goal programming problem using linear programming
[x, fval] = linprog(f, [], [], Aeq, beq, lb, ub);

% Extract solution and deviations
x_solution = x(1:2);  % Decision variables (x1, x2)
deviations = x(3:end);  % Deviation variables (d1-, d2-, d3-,
d4-, d5+, d6+)

% Display the solution
disp('Optimal Production Levels:');
disp(['Bread: ', num2str(x_solution(1))]);
disp(['Buns: ', num2str(x_solution(2))]);

disp('Deviations from Goals:');
disp(deviations);
```

solution

```
Optimization terminated.
Optimal Production Levels:
Bread: 1.3634e-07
Buns: 113235.2941
Deviations from Goals:
   1.0e+05 *


    2.6765
    0.0000
    0.0000
    0.0000
    0.0001
    0.0000
    0.0002
    0.0000
```

- ✓ No.of breads should be manufactured during a week: 0
- ✓ No.of buns should be manufactured during a week: 113235

Conclusion

According to the solution we get from the matlab,

The management should stop the making bread and increase the production of buns per week to maximize the company profit.

- **We suggest to the stop the production of bread and to make 113235 buns per week.**
- **According to our calculation company should make nearly 16176 buns per day to fulfill their desire**.